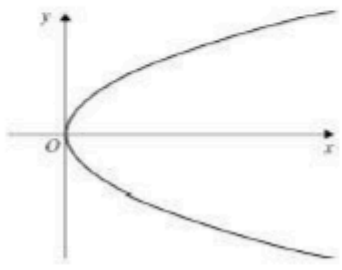


Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Sketches correct parabola	AO1.2	B1	
(b)	Obtains $\pi \int 4x \, dx$	AO3.3	M1	$\text{volume} = \pi \int_0^d y^2 \, dx$
(i)	Limits not required for this mark. Condone missing dx			$\therefore 1000 = \pi \int_0^d 4x \, dx$
	Obtains $2x^2$ and uses limits of d and 0 (oe).	AO1.1b	B1	$\frac{1000}{\pi} = \left[\frac{4x^2}{2} \right]_0^d$
	Forms an equation of the form $kd^2 = \text{volume}$ (oe)	AO3.4	M1	$\frac{1000}{\pi} = 2d^2 - 0$ $2\pi d^2 = 1000$
	Correct depth to nearest millimetre. Condone 126 or 12.6 without units. NMS: 126 or 12.6 scores 4/4. Using 1000000 mm ³ leads to a correct answer of 399 mm for 4/4.	AO3.2	A1	$d = \sqrt{\frac{500}{\pi}}$ $\text{depth} = 12.6 \text{ cm}$
(ii)	States appropriate assumption	AO3.5b	B1	The thickness of the plastic is negligible
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Obtains y^2	AO1.1b	B1	$y^2 = 9 + 6\sqrt{x} + x$
Integrates 'their' y^2 within the integral to find the volume of revolution with at least two terms correct (condone missing π)	AO1.1a	M1	$\text{Volume} = \pi \int_1^4 (9 + 6\sqrt{x} + x) \, dx$ $= \pi \left[9x + 4x^{\frac{3}{2}} + \frac{x^2}{2} \right]_1^4$



Obtains all terms correctly FT 'their' y^2 , provided M1 awarded	AO1.1b	A1F	
Substitutes correct limits into 'their' volume expression FT provided previous M1 awarded	AO1.1a	M1	substituting limits $\left[9 \times 4 + 4 \times 4^{\frac{3}{2}} + \frac{4^2}{2} \right]$ $- \left[9 \times 1 + 4 \times 1^{\frac{3}{2}} + \frac{1^2}{2} \right]$ $= 76 - 13\frac{1}{2} = 62\frac{1}{2} = \frac{125}{2}$
Completes a fully correct argument to obtain correct expression with $\pi \int y^2 dx$ found at some earlier stage in the working	AO2.1	A1	Hence volume = $\frac{125\pi}{2}$ AG
Total 5 marks			

Q3.

$$V = \pi \int x^2 dy$$

$$16x^2 - (y-8)^2 = 32$$

$$x^2 = \frac{1}{16}(y-8)^2 + 2$$

OE

B1

$$V = (\pi) \int_{(0)}^{(16)} \left(\frac{1}{16}(y-8)^2 + 2 \right) (dy)$$

Accept 'their' x^2 in terms of y Condone missing limits and π **wherever bracketed**

M1

$$V = (\pi) \left[\frac{1}{16} \times \frac{1}{3} (y-8)^3 + 2y \right]_{(0)}^{(16)}$$

OE, for correct integration of correct integrand

A1

$$V = (\pi) \left[\frac{1}{16} \times \frac{1}{3} (16-8)^3 + 2(16) - \frac{1}{16} \times \frac{1}{3} (-8)^3 \right]$$

OE, correct use of correct limits in correct expression,

PI by correct answer.

$$V = \frac{160}{3} \pi$$

A1

OE exact value, eg $\pi 53 \frac{1}{3}$ or $\pi 53.3$ or $\frac{2560}{48} \pi$

A1

[5]

Q4.

(a) $\left[\frac{dy}{dx} = \right] p e^{-\frac{1}{4}x} x^2 + q x e^{-\frac{1}{4}x}$

p, q constants

M1

$$p = -\frac{1}{4} \text{ and } q = 2$$

A1

$$\left[\Rightarrow e^{-\frac{1}{4}x} \left(-\frac{1}{4}x^2 + 2x \right) = 0 \right]$$

$$e^{-\frac{1}{4}x} \neq 0$$

or $e^{-\frac{1}{4}x} = 0$ impossible OE (may be seen later)

E1

$$\left(e^{-\frac{1}{4}x} \right) (ax^2 + bx) = 0$$

$$\text{or } e^{-\frac{1}{4}x} x(ax + b) = 0$$

m1

$$x = 0, 8$$

A1

$$x = 0, y = 0$$

A1

$$x = 8, y = 64e^{-2}$$

*condone $y = 82 e^{-\frac{8}{4}}$ etc
ignore further numerical evaluation*

(b) (i) $\int x^2 e^{-\frac{1}{4}x} dx$ $u = x^2$ $\frac{dv}{dx} = e^{-\frac{1}{4}x}$ $\frac{du}{dx} = 2x$ $v = k e^{-\frac{1}{4}x}$
where k is a constant

M1

$$k = -4$$

A1

$-4x e^{-\frac{1}{4}x} - \int -4e^{-\frac{1}{4}x} \times 2x dx$, or better
correct substitution of their terms

A1F

$u = mx$ $\frac{dv}{dx} = ne^{-\frac{1}{4}x}$ $\frac{du}{dx} = m$ $v = -4n e^{-\frac{1}{4}x}$
both differentiation and integration must be correct

m1

$$\int = -4x^2 e^{-\frac{1}{4}x} + 8 \left(-4x e^{-\frac{1}{4}x} + \int 4e^{-\frac{1}{4}x} dx \right)$$

$$= \left[-4x^2 e^{-\frac{1}{4}x} - 32x e^{-\frac{1}{4}x} - 128 e^{-\frac{1}{4}x} \right]_{(0)}^{(4)}$$

A1

$$= -e^{-1}[64 + 256] - [-128]$$

correct substitution and attempt at subtraction

in $ax^2 e^{-\frac{1}{4}x} + bx e^{-\frac{1}{4}x} + ce^{-\frac{1}{4}x}$ (may be in 3 stages)

m1 (dep on M1 only)

$$= 128 - \frac{320}{e}$$

or $128 - 320e^{-1}$ ignore further numerical evaluation

A1

(ii) $v = \pi \int_{(0)}^{(4)} 9x^2 e^{-\frac{1}{4}x} dx$

condone omission of brackets, limits

M1



$$= 9\pi \left(128 - \frac{320}{e} \right)$$

$9\pi \times (\text{their exact } b(i))$

A1F

2

[16]

Q5.

(a) $\int x e^{6x} dx$

$$\left. \begin{array}{l} u = x \\ \frac{du}{dx} = 1 \end{array} \right\} \begin{array}{l} \frac{dv}{dx} = e^{6x} \\ v = \frac{1}{6} e^{6x} \end{array}$$

All 4 terms in this form, $k = \frac{1}{6}, 1$ or 6

M1

$$k = \frac{1}{6}$$

A1

$$\frac{1}{6} x e^{6x} - \int \frac{1}{6} e^{6x} (dx)$$

Correct substitution of their terms into parts formula

A1F

$$= \frac{1}{6} x e^{6x} - \frac{1}{36} e^{6x} (+C) \quad \text{OE}$$

No ISW for incorrect simplification

A1

4

(b) $(V =) \int_0^1 x e^{6x} dx$

Must include π , limits and dx

B1

$$= (\pi) \left[\left(\frac{1}{6} e^6 - \frac{1}{36} e^6 \right) - \left(-\frac{1}{36} \right) \right]$$

Correct substitution of 0 and 1 into their answer in (a), must be of the form $axe^{6x} - be^{6x}$, where $a > 0$, $b > 0$ and $F(1) - F(0)$ seen

M1

$$= \pi \left[\frac{5}{36} e^6 + \frac{1}{36} \right]$$

CAO; ISW

A1

3

[7]

Q6.

$$V = \pi \int_0^{\ln 2} (4e^{-2x} - e^{-4x})^2 dx$$

Must be completely correct including dx seen on this line or next line

Limits, brackets and π PI from later working

B1

$$= (\pi) \int 16e^{-4x} + e^{-8x} - 8e^{-6x} (dx)$$

Correct expansion, PI from later working

B1

$$= (\pi) \left[-4e^{-4x} - \frac{1}{8}e^{-8x} + \frac{4e^{-6x}}{3} \right]_{(0)}^{(\ln 2)}$$

$$\frac{16}{-4}e^{-4x} \text{ OE}$$

B1

$$-\frac{1}{8}e^{-8x} \text{ OE}$$

B1

$$\frac{-8}{-6}e^{-6x} \text{ OE may be two separate terms}$$

B1

$$= (\pi) \left[\left(-4e^{-4\ln 2} - \frac{1}{8}e^{-8\ln 2} + \frac{4}{3}e^{-6\ln 2} \right) \right]$$

$$- \left(-4e^0 - \frac{1}{8}e^0 + \frac{4}{3}e^0 \right)$$

*Correct substitution of $x = \ln 2$ and 0 into their integrated expression
(must be of form $ae^{-4x} + be^{-6x} + ce^{-8x}$) and subtracting.*

PI

$$= \frac{5247}{2048} \pi$$

OE exact fraction eg $\frac{251856}{98304} \pi$

A1

[7]

Q7.

(a) $\int x \ln x \, dx$

$$\left. \begin{aligned} u &= \ln x & \frac{dv}{(dx)} &= x \\ \frac{du}{(dx)} &= \frac{1}{x} & v &= \frac{x^2}{2} \end{aligned} \right\}$$

correct direction and sign of $\frac{1}{x} \cdot \frac{x^2}{2}$

M1

$$\int = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \times \frac{1}{x} (dx)$$

A1

$$= \frac{x^2}{2} \ln x - \frac{x^2}{4} (+c)$$

A1

3

(b) $y = (\ln x)^2$

$$\left(\frac{dy}{dx} \right) = 2 \ln x \times \frac{1}{x}$$

$$\frac{k}{x} \ln x \text{ where } k = \frac{1}{2}, 1 \text{ or } 2$$

M1

$$k = 2$$

A1

2

(c) $y = \sqrt{x} \ln x$

$$V = \pi \int_1^e x (\ln x)^2 dx$$

all correct, incl brackets, π , limits and dx
(but dx may be seen BEFORE this line)

B1

$$\left. \begin{aligned} u &= (\ln x)^2 & \frac{dv}{dx} &= x \\ \frac{du}{dx} &= 2 \ln x \frac{1}{x} & v &= \frac{x^2}{2} \end{aligned} \right\}$$

correct direction with $\frac{du}{dx} = \frac{k}{x} \ln x$ where
 $k = \frac{1}{2}, 1$ or 2 and sign of $\frac{x^2}{2}$

M1

$$\int = \frac{x^2}{2} (\ln x)^2 - \int \frac{x^2}{2} \times \frac{2}{x} \ln x (dx)$$

correct substitution of their terms into the
parts formula

m1

$$= \frac{x^2}{2} (\ln x)^2 - \int x \ln x (dx)$$

integral needs to be simplified to $\int x \ln x$

A1

$$= \frac{x^2}{2} (\ln x)^2 - \frac{1}{4} x^2 (2 \ln x - 1) \quad \text{OE}$$

$$V = \pi \left[\frac{x^2}{2} (\ln x)^2 - \frac{1}{4} x^2 (2 \ln x - 1) \right]_1^e$$

$$= \pi \left[\left(\frac{e^2}{2} - \frac{1}{4} e^2 \right) - \left(0 + \frac{1}{4} \right) \right]$$

correct substitution of 1 and e into their expressions of
the form $px^2 (\ln x)^2 + qx^2 \ln x + rx^2$ where p, q and r are
non-zero rational numbers, and an intention to subtract
Do not condone $F(1) - F(e)$

m1



$$= \frac{\pi}{4} [e^2 - 1]$$

OE

$$\pi \left[\frac{e^2}{4} - \frac{1}{4} \right] \text{ etc}$$

A1

6

[11]

Q8.

(a)

x	y
1.5	1.98100
4.5	3.22883
7.5	4.11496
10.5	4.74710

 x values correct PI

B1

3 + y values correct to 2sf or better or exact values

M1

1.981, 3.228 / 9, 4.114 / 5, 4.747 for y
(or better)

A1

$$\int = 3 \times \sum y = 42.2$$

(Note: 42.2 with evidence of mid-ordinate rule with four strips scores 4/4)

A1

4

(b) (i) $y = \ln(x^2 + 5)$
 $ey = x^2 + 5$
 OE

$$x^2 = ey - 5$$

AG Must see middle line, and no errors

B1

1

(ii) $(\pi) \int (e^y - 5)(dy)$

Condone omission of brackets around $f(y)$ throughout

M1

$$= (\pi) [e^y - 5y]_{5}^{10}$$

A1

$$= (\pi) [(e^{10} - 50) - (e^5 - 25)]$$

$$F(10) - F(5)$$

m1

$$V = \pi[e^{10} - e^5 - 25]$$

CSO

including correct notation – must see dy
ISW if evaluated

A1

4

[9]

Q9.

(a) (i)

$$\int x \cos 4x \, dx \quad u = x \quad \frac{dv}{dx} = \cos 4x$$

$$\int \cos 4x, \frac{d}{dx}(x) \quad \text{attempted}$$

M1

$$\frac{du}{dx} = 1 \quad v = \frac{\sin 4x}{4}$$

All correct

A1

$$\int = x \frac{\sin 4x}{4} - \int \frac{\sin 4x}{4} \, dx$$

Correct substitution of their terms into
parts formula

m1

$$= \frac{x \sin 4x}{4} + \frac{\cos 4x}{16} (+c)$$

OE with fractions unsimplified

A1

4

(ii) $\int x^2 \sin 4x \, dx \quad u = x^2 \quad \frac{dv}{dx} = \sin 4x$



$$\frac{dw}{dx} = 2x \quad v = -\frac{\cos 4x}{4}$$

$$\int \sin 4x, \frac{d}{dx}(x^2) \quad \text{attempted}$$

M1

$$\int = \frac{-x^2 \cos 4x}{4} - \int \frac{-2x \cos 4x}{4} dx$$

A1

$$= \frac{-x^2 \cos 4x}{4} + \frac{1}{2} \int x \cos 4x dx$$

$$= \frac{-x^2 \cos 4x}{4} + \frac{1}{2} \left[\frac{x \sin 4x}{4} + \frac{\cos 4x}{16} \right]$$

Clear attempt to replace integral using their answer from part (a)(i)

m1

$$= \frac{-x^2 \cos 4x}{4} + \frac{x \sin 4x}{8} + \frac{\cos 4x}{32} (+c)$$

OE with fractions unsimplified

A1

4

$$(b) \quad V = (\pi) \int_{(0)}^{(0.2)} (64)x^2 \sin 4x (dx)$$

M1

$$= (\pi \times 64) \left[\frac{-x^2 \cos 4x}{4} + \frac{x \sin 4x}{8} + \frac{\cos 4x}{32} \right]$$

$$= \pi[2.09529 - 2]$$

Must see evidence of their (a)(ii) result or starting again obtaining 3 terms of the form $\pm Ax^2 \cos 4x \pm Bx \sin 4x \pm C \cos 4x$ AND $F(0.2) - F(0)$ attempted

m1

$$= 0.299 \quad \text{AWRT}$$

Accept AWRT 0.0953 π

A1

3

Q10.

$$V = (\pi) \int y^2 \, dx$$

$$= (\pi) \int (x-2)^5 \, dx$$

M1

$$= (\pi) \left[\frac{(x-2)^6}{6} \right]_3^4$$

limits not required

A1

$$= (\pi) \left(\frac{2^6}{6} - \frac{1}{6} \right)$$

correct substitution into $(\pi) k (x-2)^6$

m1

$$= 10.5\pi$$

*allow equivalent fraction $\left(\frac{63}{6}\pi \text{ etc}\right)$
(AWRT 10.5 or 10.5π m1, A0)*

A1

[4]

Q11.

(a) $V = \pi \int x^2 \, dx$

PI

B1

$$V = \frac{(\pi)}{4} \int (100 - y^2) \, dy$$

$k \int (100 - y^2) \, dy$ may be recovered

Allow $\int (\text{their } x)^2 \, dx$, expanded

M1

$$= \frac{(\pi)}{4} \left[100y - \frac{y^3}{3} \right]_{(0)}^{(10)}$$

A1

$$= \frac{(\pi)}{4} \left[\frac{2000}{3} \right]$$

For $F(10) - F(0)$

M1

$$= \frac{500\pi}{3}$$

OE CSO

$$\begin{aligned}
 & \left[\begin{array}{l}
 \text{SC: if rotated about } x\text{-axis} \\
 V = \pi \left[100x - \frac{4x^3}{3} \right]_0^5 \quad \text{M1} \\
 = \frac{1000}{3} \pi \quad \text{A1 max 2/5}
 \end{array} \right]
 \end{aligned}$$

A1

5

(b)

x	y	
0.5	9.95(0)	} or better
1.5	9.539	
2.5	8.66(0)	
3.5	7.141	
4.5	4.359	

Correct X

B1

4 + correct y to 2sf

M1

All y correct

A1

$$A = 1 \times \sum y = 39.6$$

(39.6 scores $\frac{4}{4}$)

A1

4

[9]

Q12.

(a) $e^{-2x} = 3$
 $-2x = \ln 3$

M1

$$x = -\frac{1}{2} \ln 3$$

OE ISW

A1

2

(b) $\int x e^{-2x} dx$

$$u = x \quad \frac{dv}{dx} = e^{-2x}$$

$$\frac{du}{dx} = 1 \quad v = -\frac{1}{2} e^{-2x}$$

differentiating and integrating

M1

$$\int = -\frac{1}{2} x e^{-2x} + \int \frac{1}{2} e^{-2x} (dx)$$

correct subs of their values into parts formula

m1A1

$$= -\frac{1}{2} x e^{-2x} + \int \frac{1}{2} e^{-2x} + c$$

No further incorrect working

A1

4

(c) (i) $y = e^{-2x} + 6x$
 $\frac{dy}{dx} = -2e^{-2x} + 6 = 0$

$$e^{-2x} + 6 = 0$$

M1

$$\frac{dy}{dx} = 0 \Rightarrow -2(e^{-2x} - 3) = 0$$

$$x = -\frac{1}{2} \ln 3$$

OE

A1



$$y = 3 + 6\left(-\frac{1}{2}\ln 3\right)$$

Correct substitute of their valid x

M1

$$= 3 - 3 \ln 3$$

OE ISW

A1

4

$$(ii) \quad \frac{d^2 y}{dx^2} = 4e^{-2x} \begin{cases} = 12 \\ > 0 \end{cases}$$

Other methods need justification

Allow error in $\frac{d^2 y}{dx^2}$ or x -value, but not both

M1

$\therefore$ minimum

A1

2

$$(iii) \quad (V) = \pi \int_0^1 y^2 dx = (\pi) \int_0^1 (e^{-2x} + 6x)^2 (dx)$$

Either

M1

$$= (\pi) \int_0^1 (e^{-4x} + 12xe^{-2x} + 36x^2) dx$$

Correct expansion

B1

$$= (\pi) \left[-\frac{1}{4} e^{-4x} - 6xe^{-2x} - 3e^{-2x} + 12x^3 \right]_0^1$$

3 correct terms; '- 6', '- 3' correct or
12 $\times$ their (b)

A1

All correct

A1



$$\begin{aligned}
 &= \pi \left[\left(-\frac{1}{4}e^{-4} - 9e^{-2} + 12 \right) - \left(-\frac{1}{4} - 3 \right) \right] \\
 &= \pi \left[15\frac{1}{4} - 9e^{-2} - \frac{1}{4}e^{-4} \right] \\
 &= 44.1
 \end{aligned}$$

AWRT

B1

5

[17]

Q13.

(a) $y = (e^{3x} + 1)^{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{2} (e^{3x} + 1)^{-\frac{1}{2}} \times 3e^{3x}$$

$$\frac{1}{2} (e^{3x} + 1)^{-\frac{1}{2}}$$

$$e^{3x}$$

M1

A1

$$\frac{3}{2} \text{ (allow } \frac{1}{2} \times 3 \text{) w.n.f.e}$$

A1

$$x = \ln 2:$$

$$\frac{dy}{dx} = \frac{3}{2} (e^{\ln 8} + 1)^{-\frac{1}{2}} \times e^{\ln 8}$$

correct substitution into their $\frac{dy}{dx}$
(must use $\ln 8$ or $\ln 23$)

M1

$$= \frac{3}{2} \times \frac{1}{3} \times 8$$

$$= 4$$

A1

5

(b)

x	y
0.25	1.765(5)
0.75	3.238(5)
1.25	6.597(1)
1.75	13.84(1)

correct x values

B1

3 or 4 correct y values 4 s.f. or better

B1

$$\int = 0.5 \times \sum y \quad \text{P.1}$$

M1

$$= 12.7$$

sc 12.7 with no working $\frac{2}{4}$

A1

4

(c)
$$v = \pi \int y^2 dx$$

$$= (\pi) \int (e^{3x} + 1) (dx)$$

M1

$$= (\pi) \left[\frac{1}{3} e^{3x} + x \right]_{(0)}^{(2)}$$

$$ke^{3x} + x$$

A1

$$= (\pi) \left[\left(\frac{1}{3} e^6 + 2 \right) - \left(\frac{1}{3} e^0 + 0 \right) \right]$$

correct substitution into $f \left(\int e^{3x} \right)$

m1

$$= \pi \left[\frac{1}{3} e^6 + \frac{5}{3} \right]$$

$$\left(= \frac{\pi}{3} (e^6 + 5) \right)$$

CSO

A1

4

[13]

Q14.

(a) $\int x \sin x dx \quad u = x$

$$\frac{dv}{dx} = \sin x$$

For differentiating one term and integrating other

M1

$$\frac{du}{dx} = 1 \quad v = -\cos x$$

$$\int = -x \cos x - \int -\cos x (dx)$$

m1

For correctly substituting their terms into parts formula

A1

$$= -x \cos x + \sin x (+c)$$

CSO

A1

4

(b) $u = x^2 + 5$
 $du = 2x dx$

$$\int = \int \frac{1}{2} u^{\frac{1}{2}} (du)$$

M1

$$\int k u^{\frac{1}{2}} (du)$$

condone omission of du but M0 if dx

$$\frac{1}{2} k = \quad OE$$

A1

$$= \frac{u^{\frac{3}{2}}}{\frac{3}{2}}$$

$$Ft \int k u^{\frac{1}{2}} du$$

A1 ✓

$$= \frac{1}{3} \sqrt{(x^2 + 5)^3} (+c)$$

CSO



$$\text{SC} \quad \frac{2}{5} \sqrt{(x^2 + 5)^3} \quad \text{with no working B3}$$

A1

4

(c) $y = x^2 - 9$

$$x^2 = y + 9$$

$$V = \pi \int (y + 9) dy$$

Must have π and x^2 , condone omission of dy ,
but B0 if $d x$

B1

$$= \pi \int (y + 9) dy$$

$$= (\pi) \left[\frac{y^2}{2} + 9y \right]_1^2 \quad \text{or} \quad (\pi) \left[\frac{(y+9)^2}{2} \right]_1^2$$

$\left. \begin{array}{l} \int "their x^2" dy \text{ integrated} \\ \text{Limits 2 and 1 substituted in} \\ \text{correct order including - sign} \end{array} \right\} \pi \text{ not necessary}$

M1

$$= (\pi) \left[20 - 9\frac{1}{2} \right]$$

m1

$$= 10\frac{1}{2} \pi$$

CSO

A1

4

[12]

Q15.

(a) $4(x-1)^3$ or in expanded form
allow $-4(1-x)^3$

B1

1

(b) $V = 4(\pi) \int_2^4 (x-1)^3 dx$



$$(\pi) \int y^2 dx$$

M1

$$= \pi \left[\frac{(x-1)^4}{4} \right]_2^4$$

$k(x-1)^4$ (π) or in expanded form

M1

correct substitution of limits into $k(x-1)^4$

m1

$$= \pi(81 - 1) = 80 \pi$$

CAO

A1

4

(c) Translate

E1

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

OE

B1

Stretch (I) SF 2 (II)
for I and (II or III)

M1

// y axis (III)
for I and II and III

A1

4

[9]

Q16.

(a) $z = \frac{\sin x}{\cos x}$

$$\frac{dz}{dx} = \frac{\cos x \cos x - \sin x (-\sin x)}{\cos^2 x}$$

M1



use of quotient rule $\left(\frac{\pm \cos^2 x \pm \sin^2 x}{\cos^2 x} \right)$

A1

$$= \frac{1}{\cos^2 x}$$

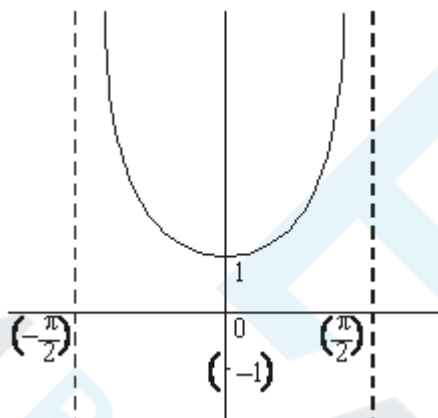
$$= \sec^2 x$$

AG (be convinced)

A1

3

(b)



correct shape including asymptotic behaviour and symmetrical about $x = 0$ and $y > 0$

use of 1

M1

A1

2

(c) $V = (k) \int \sec^2 x \, dx$

M1

$$= (k) [\tan x]_0^1$$

A1

$$= 4.89$$

CAO

A1

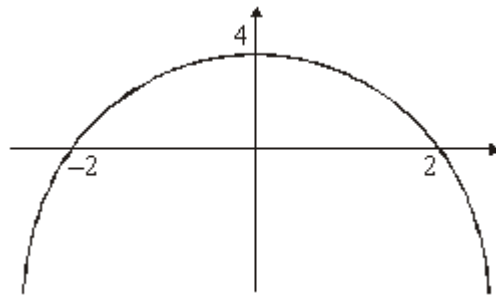
3

[8]

Q17.



(a) (i)



Shape symmetrical about y axis

all correct

M1

A1

2

(ii) $V = (k) \int (4 - x^2)^2 dx$
 $= (\pi) \int 16 - 8x^2 + x^4 dx$
expanding bracket

$$= (\pi) \left[16x - \frac{8x^3}{3} + \frac{x^5}{5} \right]$$

correctly integrating 2 of their terms

B1

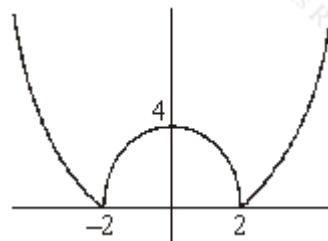
M1

$$= \pi \frac{256}{15}$$

A1

4

(b) (i)



modulus graph

shape

M1

A1

2



(ii) $|4 - x^2| = 3$

attempt at solving a correct equation

M1

$$4 - x^2 = 3 \Rightarrow x = +1, -1$$

2 correct

A1

$$4 - x^2 = -3 \Rightarrow x = \pm \sqrt{7} \text{ (or exact equivalent)}$$

2 correct

A1

3

(iii) $-\sqrt{7} < x < -1$

condone $\sqrt{7} = 2.6$ (or better)

B1F

$$1 < x < \sqrt{7}$$

B1F

2

[13]

Examiner reports

Q1.

Most students drew a correct sketch of the curve, although some drew only the top half.

In part (b), the majority of students used centimetres and produced a bowl with sensible dimensions. This question was generally well answered, although a significant minority of students initially included π , but then lost it during their calculations.

In part (c), many students gave a sensible assumption, usually referring to the thickness of the material used. Some incorrect responses questioned the capabilities of 3-D printers.

Q3.

Many students scored full marks here. Those who did not gain all the marks generally lost them at the start when they made sign errors in isolating x^2 . Other errors mainly involved the handling of the 16. An error that frequently occurred on integration was that the

constant often became a multiple of x , e.g. $\frac{\pi}{16} \left(\frac{y^3}{3} - 8y^2 + 96x \right)$; although most recovered on the next line by using the correct limits.

Q4.

In part (a), most candidates scored the first two marks, but some missed out the minus sign when differentiating. The main problem was candidates' inability to deal with $e^{-\frac{1}{4}x}$, with $\frac{-x}{4}e^{-\frac{1}{4}x}$ being the most common error. Where correct initial solutions were given, the next problem was forming a quadratic and then dealing with $e^{-\frac{1}{4}x} \neq 0$. Very few candidates obtained the E mark, and many lost the m mark for not showing a viable quadratic. Another very common error was when $x = 0$ to give the coordinates as (0, 1). A significant number of candidates failed to find the y values after an otherwise-correct solution.

In part (b)(i), the concept of integration by parts seemed to have been met by the majority of candidates and most went in the right direction; those who chose $\frac{dv}{dx}$ to be x^2 could not recover any marks, and the same was true of values of v such as $v = \frac{-4}{x}e^{-\frac{1}{4}x}$, which were often seen. Those candidates who correctly obtained $v = -4e^{-\frac{1}{4}x}$ usually obtained the first 4 marks, but there were often sign errors in the final expression, with $-4x^2e^{-\frac{1}{4}x} - 32xe^{-\frac{1}{4}x} + 128e^{-\frac{1}{4}x}$ being the most common error.

Candidates who lost marks in part (b)(i) often obtained both the marks in part (b)(ii) by following through correctly with their solution. The main error with this part was using a coefficient of 3 rather than 9 after squaring the expression for y .

Q5.

Part (a) was answered very well by many of the students, the main error being $v = 6e^{6x}$ instead of $\frac{1}{6}e^{6x}$. The other error was in the final answer, where $\frac{1}{6} \times \frac{1}{6} = \frac{1}{12}$ was a common



sight. Again many correct responses were seen in part (b), but this was the first question where many students started to lose marks. Many could not set up the initial formula for the volume, with the major error being the omission of dx . A significant number of students also failed to see the connection with part (a) and started again often getting different results to their previous response. The other main error was the substitution of the limits with $xe6x$ often being evaluated as $\frac{1}{6}$ when $x = 0$ was substituted.

Q6.

Most candidates lost the first B mark, usually for missing the ' dx '. Attempts at squaring the function often went awry; losing the middle term, losing the negative index, getting an x^2 power were all seen. After such errors only a few more marks were available. Treating the function as a square and thinking that they could integrate to a cube was very disappointing to see on an A2 paper.

Q7.

In part (a) in applying integration by parts, it was essential to start off in the correct direction and many failed at the first fence. A few fell by integrating x to x^2 and losing the $\frac{1}{2}$.

It was disappointing how many candidates failed to simplify the second integral properly and integrate it to get $\frac{x^2}{4}$.

In (b) the differentiation was generally well done, mainly using the chain rule but also the product rule.

In part (c) it should be clear from previous reports that the initial mark here requires a fully correct integral, simplified in terms of x , including dx , with the limits and in this case also π . The function needed to be squared and this proved to be the downfall of many candidates as they needed to use correct notation. Able candidates then recognised how to split the integral, apply integration by parts and use part (b) and then part (a) [or vice versa] to complete. A significant number had a sign error in their final expression as, again, they failed to use brackets correctly.

Q8.

Part (a) was very well answered by the majority of candidates. Some candidates lost a mark through not giving the answer to the required degree of accuracy. Candidates losing the final mark had sometimes lost the previous mark by showing their working to insufficient accuracy (usually 3sf). Few attempts at anything other than the mid-ordinate rule were seen.

In part (a)(i), most candidates finished with the required expression though not all had derived it through valid means.

Most candidates obtained some marks on part (a)(ii), and many fully correct responses were seen.

The final A mark was often lost because dy had been omitted. Other errors

occurred during the integration, with $e^y - 5x$ being very common and $\frac{e^y}{y} - 5y$ also being popular.

Q9.

- (a) (i) Very few candidates did not attempt this part and it was well answered with most candidates gaining at least part marks. The most common error was to get $v = 4\sin x$ rather than $\frac{1}{4}(\sin 4x)$. This still gave the possibility of earning both method marks.
- (ii) The first method mark was gained by most candidates for setting up the parts correctly. Many candidates who had trouble integrating $\cos 4x$ in part (a)(i) repeated their error here with $v = 4\cos x$ rather than $-\frac{1}{4}(\cos 4x)$ and therefore lost the A mark. Good candidates at this stage realised they could use their answer to part (a)(i) to write down the required solution but I am afraid most chose to do integration by parts twice in order to find a solution.
- (b) This was not very well answered since the relationship to part (a)(ii) often went unnoticed. Most candidates obtained the first method mark for $\int x^2 \sin 4x \, dx$. The majority of candidates started again, some of them with some success, but for most their expressions usually were integrated to $Kx^3 \sin 4x$ and no further credit was available.

Q10.

This question was very well answered by the majority of candidates with many achieving full marks. Some candidates lost the final mark for several reasons: the introduction of (+ c), the omission of π or by not giving the exact answer as required. Some candidates gained no credit as they were unable to handle the squaring of $\sqrt{(x-2)^5}$ and other candidates lost 3 marks by differentiating rather than integrating $(x-2)^5$.

Q11.

Many candidates lost marks on this question from careless work. Some candidates scored very well on this question and full marks were not uncommon amongst the more able candidates.

Many candidates obtained full marks in part (a). Where candidates scored 4 out of 5 it was generally because at no stage did they indicate ' dy ' and hence lost the B mark. A

very common error was in $\int (100 - y^2) dx$, where the result was often $(100x - y^3/3)$, again with candidates very unsure as to whether they were integrating w.r.t. x or y . Many special cases were seen and, where attempted, candidates often achieved 2 marks. Several candidates found it difficult to isolate x^2 .

Although many candidates answered part (b) correctly they often lost the final mark for not writing the answer to 3sf. A common error was also to miss out one of the x values, usually the 0.5. Candidates still seem not to understand that if they require an answer to 3sf then they should be showing either exact values of y or working to 4 sf.

Q12.

Part (a) was well answered by the majority of candidates. A common error was $x = -\ln \frac{3}{2}$.

The integration by parts in part (b) was well done, with many candidates achieving full marks. There were a few candidates who used $u = e^{-2x}$ and $dv = x$ and hence produced a more complicated expression, although there were fewer this session than in the past.

In part (c)(i), many candidates obtained the correct derivative and went on to obtain the correct x -coordinate. Unfortunately they then lost marks by not finding the appropriate y -value. Candidates who made earlier errors were usually able to obtain some credit from a correct method.

In part (c)(ii), most candidates at least obtained the method mark, and many using the correct exact value of x successfully completed the question.

In part (c)(iii), most candidates achieved the method mark and many went on to get the mark for the expansion, although e^{4x} and e^{4x^2} were common errors for the first term. The integration was poorly done with most candidates failing to realise that they could use their earlier result from part (b). Some fully correct solutions were seen.

Q13.

Most candidates produced the correct derivative in part (a). The main error was the omission of the 3. Substitution of $x = \ln 2$ was usually correct although there were several cases of $e^{3\ln 2} = 6$.

Part (b) was generally very well answered. The majority of candidates attempted the mid-ordinate rule with many fully correct responses seen. Errors occurred with working with 3 s.f. and writing the final answer to an inappropriate degree of accuracy.

Part (c) was done very well up until the last step where a very large number of candidates thought that $2 - 1/3 = -5/3$ and failed to gain the final mark.

Q14.

In part (a), many candidates made a very good attempt at integration by parts, with full

marks often being obtained. A major error was $\frac{dy}{dx} = \sin x$ leading to $x = \cos x$ however, candidates who had shown their components for using parts were still able to obtain half marks. The omission of dx was common.

In part (b), relatively few candidates were able to change the variable; many candidates were unable to handle dx and could not write the integral in terms of u and du . Where the final answer was correct, candidates had omitted du in their working, resulting in the loss of marks. However, well prepared candidates did produce fully correct solutions.

In part (c), many candidates answered this question correctly, although $\int x^2 dx$ was common as was 2π instead of π . A significant number of candidates wrote

$$\int (y+9) dy = \frac{y^2}{2} + 9x.$$

Q15.

Part (a) was very well answered by the majority of candidates. There were a great number of responses that used the substitution $u = x - 1$.

Part (b) proved difficult for many of the candidates, although fully correct responses were

seen. Most candidates gained the first M mark, although $\int x^2 dx$ was seen. Those candidates who evaluated y^2 correctly and realised the significance of part (a) went on to gain most marks, although many omitted π . Most marks were lost by candidates who

incorrectly evaluated y^2 , to involve powers of $\frac{3}{2}$, and integrated to get powers of $\frac{5}{2}$.

In part (c), most candidates gained part marks. The word 'transformation' rather than 'translation' was common. Many candidates used enlargement rather than stretch; the use

of the x -axis, rather than the y -axis, was also common, as was scale factor $\frac{1}{2}$.

Q16.

Part (a) This was well answered by most candidates with many obtaining full marks. Where errors occurred it was mainly in the numerator of the derivative with incorrect signs.

Part (b) This was poorly answered with axes often not labelled. Graphs were often

imprecise and did not show the asymptotic behaviour at $\frac{\pi}{2}$.

Part (c) Very well answered by the majority of candidates. Rounding again was an issue with a common incorrect answer of 4.90.

Q17.

Part (a)(i) was well answered although some scripts used an approach of making a table of values and drawing a graph on graph paper. Part (a)(ii) tested the knowledge and use of the formula for volume of rotation. A number of candidates did not know the correct formula and a significant number of candidates could not expand $(4 - x^2)^2$ correctly. Candidates could draw a modulus graph in part (b), but many had the wrong shape of the curve. Candidates attempted to solve the equation in part (b)(ii) with varying degrees of success. In the final part (b)(iii), successful candidates used the previous graph together with the solution of the intersection from part (b)(ii).

This question was the least well answered question on the paper with very few correct solutions. Although the graphs of inverse trigonometrical functions is stated in the specification there were many incorrect graphs sketched. Many candidates had their calculators in degree mode and in part (b)(ii) found the values of 19 and 21 and then stated, 'change of sign, therefore root'. There were similar mistakes in part (c) with iterations jumping from 0.8 to 44 with candidates apparently not realising there was something amiss.